

Term Paper Summary

Computing the Impact of Investment in Renewable Energy Projects on
Output Growth of Developing Countries – Testing for Conditional
Convergence

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Summary

This study follows the landmark study “A Contribution to the Empirics of Economic Growth” by Mankiw, Romer and Weil from 1992. This empirical paper studies the impact of investment in renewable energy on total output using an augmented Solow model. The augmented Solow’s model[1](Mankiw et al.) regards the textbook Solow’s model [2](Solow) highly. There is no consensus in the existing literature on impact of renewable energy and growth rates of output. The literature is very sensitive to the model misspecification. Chang(see *Chang et al. 2009*) has argued for the presence of a unidirectional causality between output growth and investment in energy[3]; Giraud on the other hand(see *Giraud G. & Kahraman Z. (2014)*), argues that energy consumption causes growth which validates the growth hypothesis. [?]. A more recent study conducted by Fan and Hao (see *Fan, W., Hao, Y. (2019)*) proves a modest slowdown in growth rates will generate a significant boost in renewable energy consumption conditional on targeted foreign direct investment in the long run [4].

This paper looks at whether the addition of a proxy for renewable energy investment lowers the coefficient on the initial level of output in the augmented solow’s model. That is, does it accelerate the convergence speed. It uses a cross-country panel dataset of 90 countries. The dataset is a compilation of the Penn world Table, World Bank Data and International Renewable Energy Association(IRENA) datasets. The findings of this paper are that in the short run there is no relationship between the amount of investment in renewable energy projects and output growth. The growth rate of a country is not a statistically significant determiner of the amount of investment in renewable energy projects; in the short run. Countries with lower growth rates (Growth rates which fall below the regression line, refer to ??) tend to have higher amount of investment in renewable energy projects. In the long run, there exists a statistically significant impact of investment in renewable energy projects on output growth rates. An investment in renewable energy would hence benefit developing countries more. This relation passes all robustness checks conducted – Dataset check, Year fixed effects & A change in proxy for renewable investment.

The regression model used to derive at the relationship between investment in renewable energy projects and GDP growth is as follows

$$\begin{aligned} \ln\left(\frac{GDP}{Population}\right)_{2020} - \ln\left(\frac{GDP}{Population}\right)_{1970} = & \beta_0 + \beta_1 \ln\left(\frac{GDP}{Population}\right)_{1970} + \beta_2 \ln\left(\frac{I}{GDP}\right) \\ & + \beta_3 \ln(n + \gamma + \delta) + \beta_4 \ln(PrimarySchooling) + \\ & \beta_5 \ln(I_{RE}) + \beta_6 \ln(Employment_{RE}) + \epsilon_i \end{aligned}$$

Adding controls for renewable energy investment, the coefficient reduces and the model becomes a better fit, implying that an investment in renewable energy, in the long run, will help countries reach their steady states faster. Developing countries will take a longer time to reach their steady states as compared to developed countries. An increase in the rate of convergence would speed up this process for developing countries due to increasing rate of returns. Based on the assumption that developed countries are further away from developed countries in terms of their output growth rates and would tend to experience a lower increasing rate of returns. An investment in renewable energy would hence benefit developing countries more.

	Dependent variable:	
	log(Growth rate of Expenditure in GDP in real terms(Average GDP growth rate), Source - Penn World Table 10.0')	
	(1)	(2)
log('The GDP at constant dollars in 1970(Year of reference post oil crises of 1970), Source - Penn World Table 10.0')	-0.036 [*] (0.019)	-0.026 (0.017)
log('n+gamma-delta(gamma assumed 0.02 where value is NaN)')	0.179 ^{***} (0.040)	0.176 ^{***} (0.040)
log('Average years of secondary schooling - Proxy for Human Capital Accumulation (Source: Barro, Robert and Jong-Wha Lee, 2013)')	-0.020 (0.077)	0.020 (0.068)
log('Average Public Investment In Renewable Energy Projects In US Dollars(Millions), Source - IRENA')	0.029 (0.030)	
log('Percentage of Persons Employed in Renewable Energy Sector by working age population of country, Source- IRENA(Employed) and World Bank(WAP)')	0.008 (0.024)	
log('The average share of Investment in GDP, Source - Penn World Table 10.0')	0.109 (0.104)	0.095 (0.099)
Constant	1.838 ^{***} (0.387)	1.677 ^{***} (0.290)
Observations	66	66
R ²	0.283	0.266
Adjusted R ²	0.210	0.218
Residual Std. Error	0.225 (df = 59)	0.224 (df = 61)
F Statistic	3.884 ^{***} (df = 6; 59)	5.524 ^{***} (df = 4; 61)
Note:	* p<0.1; ** p<0.05; *** p<0.01	

	Dependent variable:	
	log(Average growth rate of GDP PPP Adjusted, Source - World Bank Data)	
	(1)	(2)
log('The GDP at constant dollars in 1970(Year of reference post oil crises of 1970), Source - Penn World Table 10.0')	-0.040** (0.020)	-0.047** (0.023)
log('n+gamma+delta(gamma assumed 0.02 where value is NaN)')	0.173*** (0.048)	0.169*** (0.049)
log('Average years of secondary schooling - Proxy for Human Capital Accumulation (Source: Barro, Robert and Jong-Wha Lee, 2013)')	-0.200** (0.082)	-0.215** (0.092)
log('Average Public Investment In Renewable Energy Projects In US Dollars(Millions), Source - IRENA')		-0.016 (0.036)
log('Percentage of Persons Employed in Renewable Energy Sector by working age population of country, Source- IRENA(Employed) and World Bank(WAP)')		0.030
log('The average share of Investment in GDP, Source - Penn World Table 10.0')	0.112 (0.119)	(0.029)
Constant	2.171*** (0.348)	0.080 (0.125)
Observations	66	66
R ²	0.335	0.347
Adjusted R ²	0.291	0.281
Residual Std. Error	0.268 (df= 61)	0.270 (df= 59)
F Statistic	7.681*** (df= 4; 61)	5.233*** (df= 6; 59)
Note:	* p<0.1; ** p<0.05; *** p<0.01	

Bibliography

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